NVIDIA SENIOR DESIGN PROJECT (SDP) — AUTOMOTIVE SURVEILLANCE MODE (ASM)

End-to-End Perception Workflow: Temporal Motion Selection & Spatial DINOv2 Token Caching

Target Architecture: NVIDIA RTX PRO 4000 Blackwell (24GB VRAM) | Shared DINOv2 ViT-S/14 | Zero Retraining Required

2.00x DINOv2 SPEEDUP
129.3 ms -> 64.7 ms (4.98 ms sparse)

-41.7% PEAK VRAM REDUCTION
4,157 MB -> 2,423 MB | +79.9% FPS

STAGE 1: TEMPORAL REDUCTION

Input Conditioning & Motion-Aware Selection

1.1 Video Stream Sources

- **Option A: 15 / 30 FPS Public Datasets**
  (MEVA, VIRAT, AVA, test_clip_10s.mp4)
- **Option B: Native 5 FPS Vehicle Surround**
  (Front, Rear, Left, Right Fisheye Cameras)

Technique A: Windowed Argmax Selection

[For 15/30 FPS Input -> Strictly Capped at 5.0 FPS]

Ultra-Fast Thumbnail Differencing (< 0.05 ms)
- Resize frames to 160x160 thumbnails on GPU
- Compute frame delta: M(f) = Mean(|F_t - F_{t-1}|)

3-Frame Temporal Windows (200 ms span)
W_k = [f_{3k}, f_{3k+1}, f_{3k+2}] for k in 0..49
Selection: f*_k = argmax_{f in W_k} M(f)
Guarantees 100% Peak Motion Capture (0% Blind Spots)

Strict 5.0 FPS Enforcement Proof:
Sum(1 frame per W_k) = 50 frames / 10s = 5.0 FPS
+24.8% motion captured | Recovered 13 boxes (279 total)

Technique B: Intelligent Idle Sentry Gating

[Pradipt's Note: Skipping within 5 FPS Stream]

Lightweight Activity Detection (Δ < 0.05 ms)
Computes global active patch fraction Δ
Threshold Δ = 2.0% (Derived from sensor SNR floor)

Dual-Branch Decision Gate:
- **IF Δ < 2.0% (Empty Street / Zero Motion):**
  -> SLEEP: Bypass DINOv2 & YOLO completely
  -> Retain prior bboxes in memory (~99% power save)
- **IF Δ >= 2.0% (Pedestrian / Vehicle Entering):**
  -> WAKEUP: Forward frame to Spatial Token Caching

Physical SNR Justification for 2.0%:
- 2.0% of 2,025 patches = ~40 patches
- Sensor thermal noise = 0.3% - 0.8% (<16 patches)
- Distant pedestrian + 5x5 dilation = 35-45 patches

STAGE 2 & 3: SPATIAL TOKEN CACHING & DINOv2

Patch Gating Engine, Sparse ViT Execution, & GPU Cache Refill

2.1 Frame Standardization (640x640 Resolution)

- Input normalized to ImageNet [0, 1] tensor on CUDA: F_t in R^{1 x 3 x 640 x 640}
- Cropped to 630x630 active patch grid: 45 x 45 = 2,025 Spatial Patches (14x14 px)
- **Empirical Sentry Redundancy: 91.97% static background | 8.03% foreground active**
- Cosine similarity between consecutive frames = 0.9772 (Extreme temporal stability)

2.2 GPU Patch Difference Detector (< 0.05 ms)

Ultra-fast PyTorch CUDA Kernel using 2D Average Pooling

Step 1: Patch MAE Computation via F.avg_pool2d
Δ_pixel = Mean_c(|F_t - F_{t-1}|) -> diff in R^{1 x 1 x 630 x 630}
patch_diff = F.avg_pool2d(diff, k=14, stride=14) in R^{1 x 1 x 45 x 45}
Binary threshold: mask = (patch_diff > τ) with τ = 0.015

Step 2: 5x5 Morphological Dilation Padding
mask_dilated = F.max_pool2d(mask, k=5, stride=1, pad=2)
Expands active boundaries by 2 patches in all directions
Guarantees object edges and boundary context are 100% captured

2.3 Adaptive Routing & Keyframe Management

KEYFRAME (20%)
- Trigger: Period K=5 (1.0s) OR Large cut (Δ > 40%)
- Evaluates all 2,026 tokens
- Fully refreshes GPU cache
- **Latency: 25.49 ms / frame**

SPARSE FRAME (80%)
- Trigger: Normal sentry (Δ <= 40%)
- Forwards active + CLS only
- Sequence length: ~163 tokens
- Attention FLOPs cut > 100x
- **Latency: 4.98 ms / frame!**

Stage 3: DINOv2 ViT Sparse Forward & GPU Buffer

Sparse ViT Encoder Slicing (12 Transformer Blocks):
- ViT self-attention natively handles arbitrary sequence lengths N x D
- Positional embeddings are injected at layer 0 (positions preserved)
- Tokens forwarded: T_sparse = [CLS, Patch_active_1, ..., Patch_active_M]
- **Mean evaluated sequence: 876 tokens across stream (-56.7% tokens)**

In-Place GPU Feature Cache Refill [Layers 3, 6, 9, 12]:
Cache Buffer: C_layer in R^{1 x 2026 x 384} allocated once in GPU VRAM
In-Place Overwrite: C_layer[:, active_indices, :] = sparse_output
Result: Produces complete 45x45 feature maps without computing background!

STAGE 4: PERCEPTION HEADS & OUTPUT

Downstream Task Execution with Zero Retraining

4.1 Object Detection Head

ViTYOLOAdapter_v2 + YOLO26s (NMS-Free)

Feature Pyramid Construction:
- Consumes cached layers [3, 6, 9, 12] reshaped to 45x45
- Multi-scale pyramid: P3/8, P4/16, P5/32 scales
- Detail Branch: CNN on raw frames preserves fine edges

YOLO26s Detection & Accuracy Preservation:
- End-to-end NMS-free inference via dual-head matching
- **266 / 277 boxes detected (96.0% overall retention)**
- **0.968 Mean BBox IoU match against Golden Baseline**
- Cars: 95.9% | Persons: 97.7% | Motorcycles: 100%+

Zero Retraining Contract:
Official adapter_best.pt & yolo26s.pt weights remain 100% untouched

4.2 Action Recognition Head

Whole-Scene Kinetics-400 Linear Probe

Temporal CLS Token Aggregation:
- **[CLS] token is actively forwarded on EVERY frame**
- Sequence of CLS tokens in R^{T x 384} across time
- Preserves unbroken 8-frame sliding temporal window

Linear Probe Classification (assets/k400_outputs):
- Temporal mean pooling over 384-dim CLS sequence
- Top-1 Action Baseline: 'playing badminton' (42.1%)
- Top-1 Action Sparse: 'playing badminton' (41.8%)
- **Agreement: 100% exact match across all temporal windows**

Zero Action Degradation:
Active CLS forward guarantees temporal action awareness is unbroken

4.3 Unified Output Coordinator

- perception.json: Bounding boxes + action predictions per frame
- detection_annotated.mp4: Overlay with YOLO26 bounding boxes
- action_annotated.mp4: Overlay with action recognition banners
- annotated_grid.mp4: Synchronized 2x2 multi-camera stitched grid
- **Sentry Security Alerting: Dispatches instant alert on perimeter breach**
- Hardware Telemetry: Active tokens, latency (ms), and VRAM logs
- **Effective Throughput: 10.38 FPS (+79.9% headroom on Blackwell)**
- **VRAM Footprint: 2,423 MB (-41.7% memory savings on GPU)**

KEY ARCHITECTURAL HIGHLIGHTS FOR NVIDIA SPONSORS & EVALUATORS:

1. Temporal Optimization: Strictly enforces 5.0 FPS budget via Windowed Argmax (0% blind spots) and skips idle empty frames via SNR 2.0% gating (<0.05 ms).
2. Spatial Token Reduction: Reduces evaluated DINOv2 tokens by 56.7%, dropping sparse ViT latency to 4.98 ms and peak VRAM by 41.7% on Blackwell GPU.
3. Downstream Compatibility: In-place GPU cache re-filling guarantees 100% plug-and-play operation with Adapter-v2 and YOLO26s (zero weight modification).
4. Perception Integrity: Preserves 96.0% object bounding boxes (0.968 IoU) and achieves 100% agreement on temporal action recognition.